CCC-C-443b
March 28, 1963

SUPERSEDING
Fed. Spec. CCC-C-443a
October 20, 1960

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK; (SINGLE AND PLIED FILLING YARNS, FLAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for unbleached flat duck.

1.2 Classification.

1.2.1 *Types.* The cloth, cotton, flat duck shall be of the following types as specified (see 6.2):

Type I.—Single filling.

Type II.—Double filling (plied).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.

CCC-T-191—Textile Test Methods.

PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific

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TABLE I.- Physical Requirements, type I, Single Filling

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength 2/ pounds (minimum)		Air permeability (cubic ft/min/ft 2/) average (maximum)	Weave
Ounces	Warp 1/	Filling	Warp	Filling		
7.26	74	27	90	55	—	Oxford, 2 ends weaving as one (plain weave)
8.47	74	27	100	65	—	
9.68	74	27	130	85	—	
10.89	74	26	135	95	—	
12.10	72	26	160	105	—	
13.32	72	26	165	110	—	
14.53	72	25	170	120	—	
14.90	76	28	180	160	—	
17.55	86	27	270	245	4	
18.15	72	25	180	140	3	

1/ Two yarns woven as one.

2/ A minus tolerance of 5 percent in breaking strength will be permitted in any one direction provided the combined strength equals the sum of the specified warp and filling requirements.

TABLE II.- Physical requirements, type II, double filling (plied)

Weight per square yard (minimum)	Yarns per inch (minimum)		Ply	Breaking strength 2/ pounds (minimum)		Weave
Ounces	Warp 1/	Filling	Filling	Warp	Filling	
7.26	86	30	2	100	60	Oxford, 2 ends weaving as one (plain weave)
8.47	86	30	2	115	85	
9.68	86	30	2	125	100	
10.89	84	30	3	140	110	
12.10	84	30	3	150	115	
13.32	84	29	4	165	135	
14.53	82	29	4	180	160	
18.15	82	25	5	195	190	

1/ Two yarns woven as one.

2/ A minus tolerance of 5 percent in breaking strength will be permitted in any one direction provided the combined strength equals the sum of the specified warp and filling requirements.

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procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication of the issue in effect on date of invitation for bids forms a part of this specification:

Textile Fiber Products Identification Act

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Material

3.1.1 *Yarns.* Yarn shall be made from cotton which has been cleaned, carded, drawn and spun into single yarn for both warp and filling for type I, and single yarn for warp, and plied yarns for filling for type II.

3.2 Physical requirements. The cloth covered by this specification shall be as specified (see 6.2) and shall conform to the requirements specified in tables I and II when tested as specified in 4.4.

3.3 Width. The width, inclusive of selvage, shall be as specified (see 6.2). A tolerance of minus 1/4 inch will be allowed on the specified width.

3.4 Nonfibrous materials. The total sizing, finishing, and other nonfibrous materials content of the finished cloth shall not exceed 12 percent when tested as specified in 4.4.

3.5 Length.

3.5.1 *Put-up.* The cloth shall be put up in rolls in accordance with PPP-P-51. Pieces 80 yards or more in length shall be delivered. The pieces shall be sewn together with a butted FSa-1 type seam on all cloth 14.90 ounces per square yard and lighter in weight, using 10/3 cord unbleached cotton thread, conforming to type IIIA of V-T-276, and shall have 6 to 8 stitches per inch. The stitching shall be placed 1/2 ± 1/16 inch from the cut or raw edge of the cloth. On cloth heavier in weight than 14.90 ounces per square yard, the pieces will be lapped and not sewn together. The number of pieces in the roll will be indicated by stencil marking on the last 6 inches of the roll on the outer surface (see 6.2).

3.5.2 *Short length.* Pieces less than 80 yards will be acceptable up to 10 percent of contract quantity, provided no piece is less than 20 yards in length, and not more than two pieces less than 80 yards are included in any one roll, except where otherwise specified in the contract (see 6.2).

3.6 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.7 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the point level specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 **Inspection for acceptance.** Except when otherwise specified by civil Government agencies, inspection shall be in accordance with the provisions of MIL-STD-105, and 4.2.1 through 4.2.3.

4.2.1 *Yard by yard examination.* The required yardage of each roll shall be examined on the face side and all defects as defined in Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except that spots, stains and streaks shall be counted regardless of cause, (i.e., not restricted to rust, oil, or grease) and only those slubs and knots which exceed the maximum acceptable limits shown for each shall be scored (see 4.2.1.2). Defects on the selvage shall not be scored unless the body of the cloth is affected. No yard shall be penalized more than four points. For purposes of this examination, a yard shall begin at the last scored defect. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 rolls taken from 25 containers. An approximately equal number of yards shall be examined in each roll of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 40.00.

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{Contracted width of material (inches)}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	—one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	—two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	—three points
For defects exceeding 9 inches in any dimension	—four points

NOTE: The following defects when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Objectionable odor.
Width—Less than minimum specified.
Excessive neppiness.

4.2.1.2 *Limits of slubs and knots.* Only slubs and knots which exceed the limits shown on figure 1 shall be scored.

4.2.2 *Examination for length.*

4.2.2.1 *Individual rolls.* During the yard by yard examination each roll in the sample shall be examined for put-up and length. Each defect shall be counted not more than once in each roll examined. The lot shall be unacceptable if two or more of the following defects are present:

DEFECTS

Any gross length found to be more than 2 yards below the gross length marked on the roll ticket.

Length of piece:

Regular lengths—Any piece less than 80 yards in length.

Short lengths—Any piece less than 20 yards in length or more than two pieces less than 80 yards in length in one roll.

For duck heavier in weight than 14.90 ounces per square yard, sewn instead of lapped.

For duck 14.90 ounces per square yard and lighter in weight:

Pieces not sewn together with a butted FSa-1 type seam.

6 to 8 stitches per inch not used in seam.

Stitching not 1/2 ± 1/16 inch from the cut or raw edge of the cloth.

Number of pieces in roll not stenciled on last 6 inches of roll on outer surface.

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4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.3 *Examination for compliance with Textile Fiber Products Identification Act.* During the yard by yard examination, each roll in the sample shall be examined for compliance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 **Examination of preparation for delivery requirements.** An examination shall be made to determine that packaging, packing and marking requirements of the applicable specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping roll fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping rolls fully prepared for delivery. The lot size shall be the number of shipping rolls in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking	Omitted, incorrect; illegible, or improper size, location, sequence or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of rolls, inadequate securing or sealing of wrapping materials, insecure sewing or loose wire ties.
Weight	Gross weight beyond the limits specified.

4.4 **Testing of the end item.** The methods of testing specified in CCC-T-191 wherever applicable and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1-1/2 yards full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size in yards	Sample size
Less than 800	2
801 to 22,000	3
22,001 and over	5

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 **Packaging.** Packaging shall be level A, B or C, as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cloth put up as specified in 3.5.1 shall be packaged in accordance with the applicable requirements for class 4 fabrics of PPP-P-51.

5.2 **Packing.** Packing shall be level A, B or C, as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirement for class 4 fabrics of PPP-P-51.

5.3 **Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-51.

6. NOTES

6.1 **Intended use.** The cloth, cotton duck, single and plied filling yarns (flat) covered by this specification is utilized in the fabrication of tentage and related items. The type I, 14.90 and 17.55 ounce per square yard flat ducks are specifically designed as emergency substitutes for type III, 12.29 ounce

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TABLE III.- Test Methods

Characteristic	Requirement Reference	Test Method
Weight	Tables I and II	5041 1/
Weave	Tables I and II	Visual 2/
Breaking strength	Tables I and II	5100
Yarn ply, types I and II	3.1.1 and Table II	Visual 2/
Yarns per inch	Tables I and II	5050
Non fibrous materials	3.4	2611
Air permeability	Table I	5450

1/ Except that the average of 5 determinations per unit sample shall be reported to the nearest 0.01 ounces.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

Army duck and Type I, No. 8 duck respectively, covered by Federal Specification CCC-C-419.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and weight of duck required (see 1.2 and 3.2).

(*c*) Width of duck required (see 3.3).

(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(*e*) Whether special marking is required (see 5.3).

(*f*) Whether short pieces will be acceptable (see 6.3).

(*g*) Provisions for short length (see 3.5).

6.2.1 In the preparation of contracts or orders it must be noted that various options, choices and alternatives as indicated in PPP-P-51 may be exercised in the preparation for delivery of cloth.

6.3 **Short pieces.** Pieces less than 80 yards in length in excess of the amount specified in 3.5 and pieces shorter in length than 20 yards may be presented to the contracting officer for consideration for end users where short length pieces may be considered.

6.4 **Color and treatment.**

6.4.1 Cloth, cotton, duck to be fire, water, weather and mildew resistant treated should be treated under the provision of Federal Specification CCC-C-428.

6.4.2 Cloth, cotton, duck to be aftertreated in any of the following conditions should be treated under the provisions of Federal Specification CCC-D-950:

(*a*) Bleached.

(*b*) Dyed.

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(*c*) Water repellent aftertreated.

(*d*) Mildew resistant aftertreated.

(*e*) Water repellent and mildew resistant aftertreated.

6.5 The air permeability for type 1 cloths, 17.55 and 18.15 ounces per square yard is to insure a tight cloth. The end uses for this type are as substitutes for tent walls, tent covers and sod cloths in time of an emergency. The other type 1 cloths are used for equipage items where air permeability is not a prime requisite.

6.6 Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

6.7 **Transportation description.** The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods, finished, not otherwise indexed by name.

Carload minimum weight—Any quantity.

Motor:

Cotton piece goods, not otherwise indexed.

Motor volume minimum weight—Any quantity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder of any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

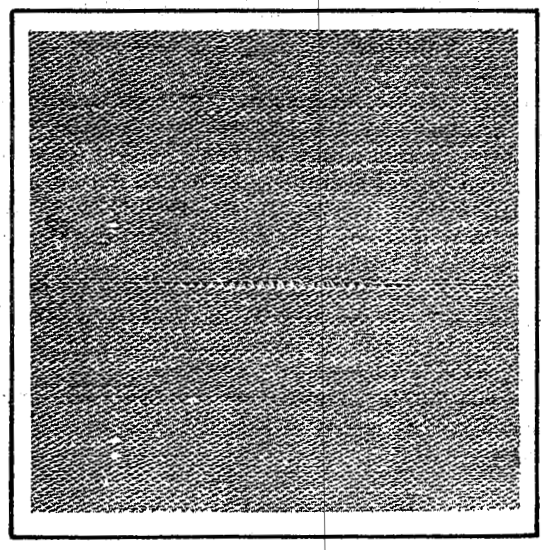
Army—GL

Navy—S

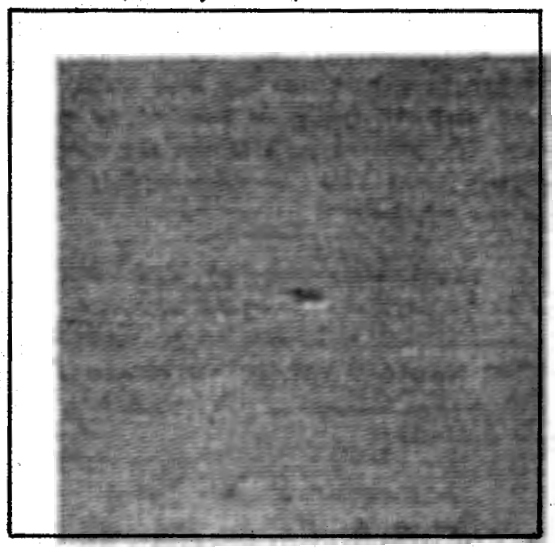
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Figure 1 (Limits of Acceptability)

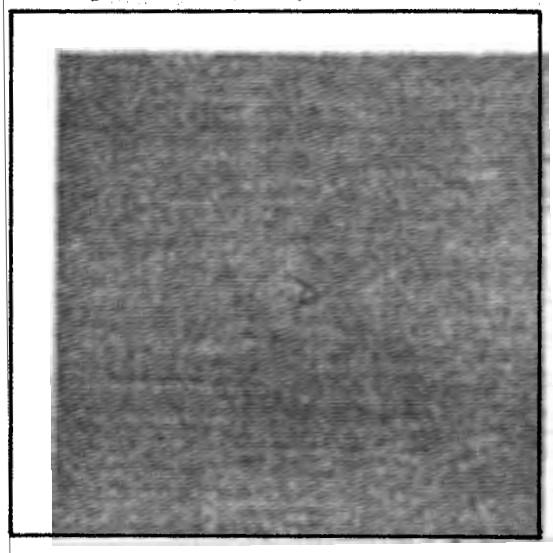
Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)



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